

Tut 5a–5c: Multiple Integrals (3 hours)

UECM1304 Calculus II

Remark: The answers are for you to check your working. There may be errors. Please report to the lecturer if you found any errors.

Double Integrals

5a(i) Double Integrals over Rectangles

- (a) Estimate the volume of the solid that lies below the surface $z = x + 2y^2$ and above the rectangle $R = [0, 2] \times [0, 4]$. Use a Riemann sum with $m = n = 2$ and choose the sample point to be lower right corners. [Ans: 36]
(b) Use the Midpoint Rule to estimate the integral in part (a). [Ans: 88]

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(define (riemann-sum-lower func xlower xupper ylower yupper Nx Ny)
  ;; Mid-point & lower differ by 'a_n'
  (define (a_n a0 h range) ; a_n = a_0 + n*h, n=0,...,n-1
    (vector-map (lambda (n) (+ a0 (* n h))) range))
  (let ((dx (/ (- xupper xlower) Nx)) (x0 xlower)
        (dy (/ (- yupper ylower) Ny)) (y0 ylower)
        (integral-val 0))
    (let ((xs (a_n x0 dx (list->vector (iota Nx))))
          (ys (a_n y0 dy (list->vector (iota Ny)))))
      (do ((i 0 (+ i 1)))
          ((>= i Nx))
        (do ((j 0 (+ j 1)))
            ((>= j Ny))
          (let*
              ((xi (vector-ref xs i)) (yj (vector-ref ys j))
               (fij (func xi yj)))
            (set! integral-val (+ integral-val (* fij dx dy)))
            (format #t "(~a,~a): f(~a,~a)=~a~%" i j xi yj fij))))
      integral-val)))
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- Evaluate the double integral by first identifying it as the volume of a solid.

(a) $\iint_R (5 - x) \, dA$, $R = \{(x, y) : 0 \leq x \leq 5, 0 \leq y \leq 3\}$ [Ans: 37.5]

(b) $\iint_R (4 - 2y) \, dA$, $R = [0, 1] \times [0, 1]$ [Ans: 3]

- Calculate the iterated integral.

(a) $\int_2^4 \int_{-1}^1 (x^2 + y^2) \, dy \, dx$ [Ans: 116/3]

(b) $\int_0^{\frac{\pi}{2}} \int_0^{\frac{\pi}{2}} \sin x \cos y \, dy \, dx$ [Ans: 1; Hint: separable]

(c) $\int_1^4 \int_1^2 \left(\frac{x}{y} + \frac{y}{x} \right) \, dy \, dx$ [Ans: $21 \ln 2/2$]

4. Calculate the double integral.

(a) $\iint_R \cos(x + 2y) \, dA$, $R = \{(x, y) : 0 \leq x \leq \pi, 0 \leq y \leq \frac{\pi}{2}\}$ [Ans: -2]

(b) $\iint_R \frac{1 + x^2}{1 + y^2} \, dA$, $R = \{(x, y) : 0 \leq x \leq 1, 0 \leq y \leq 1\}$ [Ans: $\pi/3$]

(c) $\iint_R \frac{x}{1 + xy} \, dA$, $R = [0, 1] \times [0, 1]$ [Ans: $2 \ln 2 - 1$]

(d) $\iint_R xye^{x^2y} \, dA$, $R = [0, 1] \times [0, 2]$ [Ans: $(e^2 - 3)/2$ (Hint: x then y)]

5. Find the volume of the solid that lies under the plane $3x + 2y + z = 12$ and above the rectangle $R = \{(x, y) : 0 \leq x \leq 1, -2 \leq y \leq 3\}$. [Ans: $95/2$]

6. Find the volume of the solid enclosed by the surface $z = 1 + e^x \sin y$ and the planes $x = \pm 1$, $y = 0$, $y = \pi$, and $z = 0$. [Ans: $\pi + 2e - 2$]

5a(ii) Double Integrals over General Regions

7. Evaluate the iterated integral.

(a) $\int_0^1 \int_{x^2}^x (1 + 2y) \, dydx$ [Ans: $3/10$]

(b) $\int_0^{\pi/2} \int_0^{\cos \theta} e^{\sin \theta} \, drd\theta$ [Ans: $e - 1$]

8. Evaluate the double integral.

(a) $\iint_{\mathcal{D}} \frac{4y}{x^3 + 2} \, dA$, $\mathcal{D} = \{(x, y) : 1 \leq x \leq 2, 0 \leq y \leq 2x\}$ [Ans: $8/3 \ln(10/3)$]

(b) $\iint_{\mathcal{D}} x \cos y \, dA$, $\mathcal{D}$ is bounded by $y = 0$, $y = x^2$ and $x = 1$. [Ans: $(1 - \cos 1)/2$]

(c) $\iint_{\mathcal{D}} (x + y) \, dA$, $\mathcal{D}$ is bounded by $y = \sqrt{x}$ and $y = x^2$. [Ans: $3/10$]

(d) $\iint_{\mathcal{D}} (2xy + 1) \, dA$, $\mathcal{D}$ is bounded by $y = 3x$ and $y = x^2 - 4$. [Ans: $3625/12$]

9. Find the volume of the given solid.

(a) Under the paraboloid $z = x^2 + y^2$ and above the region bounded by $y = x^2$ and $x = y^2$. [Ans: $6/35$]

(b) Enclosed by the paraboloid $z = x^2 + 3y^2$ and the planes $x = 0$, $y = 1$, $y = x$ and $z = 0$. [Ans: $5/6$]

(c) Enclosed by the cylinders $z = x^2$, $y = x^2$ and the planes $z = 0$, $y = 4$. [Ans: $128/15$ (symmetry)]

(d) Under the paraboloid $z = 2x^2 + 3y^2$ and above the region $\mathcal{D}$ in the xy -plane bounded by the line $y = |x|$ and the parabola $y = x^2$. [Ans: $29/70$ (symmetry)]

10. Sketch the region of integration and change the order of integration.

(a) $\int_0^1 \int_{4x}^4 f(x, y) \, dydx$ [Ans: $\int_0^4 \int_0^{y/4} f(x, y) \, dx dy$]

(b) $\int_0^3 \int_{-\sqrt{9-y^2}}^{\sqrt{9-y^2}} f(x, y) \, dx dy$ [Ans: $\int_{-3}^3 \int_0^{\sqrt{9-x^2}} f(x, y) \, dy dx$]

11. Evaluate the integral by reversing the order of integration.

(a) $\int_0^1 \int_{3y}^3 e^{x^2} \, dx dy$ [Ans: $(e^9 - 1)/6$]

(b) $\int_0^4 \int_{\sqrt{x}}^2 \frac{1}{y^3 + 1} \, dy dx$ [Ans: $(\ln 9)/3$]

(c) $\int_0^1 \int_x^1 e^{x/y} \, dy dx$ [Ans: $(e - 1)/2$]

5b(i) Change of Variables for Double Integrals

12. Let R be a parallelogram bounded by the lines $y = -3x$, $y = x + 2$, $y = -3x + 2$ and $y = x - 2$.

(i) A transformation T from S to R is defined by the equations

$$x = \frac{1}{4}(u + v), \quad y = \frac{1}{4}(v - 3u).$$

Find the region R and sketch the pre-image S in the uv -plane.

[Ans: Corners of R : $(1/2, -3/2)$, $(-1/2, 3/2)$, $(1, -1)$, $(0, 2)$]

(ii) Using the transformation T in part (i) or otherwise, evaluate $\iint_R (2x + 6y) \, dA$.

[Ans: 4]

13. Let R be the region in the first quadrant bounded by the lines $y = x$, $y = 2x$ and the hyperbolas $xy = 1$, $xy = 2$.

(i) A transformation T is defined by the equations $x = \frac{u}{v}$ and $y = v$. Find the image R of the region S under the transformation T . Sketch the region R in the xy -plane and the region S in the uv -plane.

[Ans: Corners of R in first quadrant: $(1, 1)$, $(\sqrt{2}, \sqrt{2})$, $(1/\sqrt{2}, \sqrt{2})$, $(1, 2)$]

(ii) Using the transformation T in part (i) or otherwise, evaluate

$$\iint_R 3xy \, dA.$$

[Ans: $9 \ln 2/4$]

14. Use $x = \frac{u}{v}$, $y = v$ or otherwise to find $\iint_R xy \, dA$, where R is the region in the first quadrant bounded by the lines $y = x$ and $y = 3x$ and the hyperbolas $xy = 1$, $xy = 3$.

[Ans: $2 \ln 3$]

15. (a) Find the image of the set S , where S is the disk given by $u^2 + v^2 \leq 1$, under the transformation $x = au$ and $y = bv$.

[Ans: ellipse]

(b) Use $x = 2u$, $y = 3v$ or otherwise to find $\iint_R x^2 \, dA$, where R is the region bounded by the ellipse $9x^2 + 4y^2 = 36$.

[Ans: $4\pi/3$]

5b(ii) Double Integrals in Polar Coordinates

16. Evaluate the given integral by changing to polar coordinates.

(a) $\iint_R (x + y) \, dA$, where R is the region that lies to the left of the y -axis between the circles $x^2 + y^2 = 1$ and $x^2 + y^2 = 4$.

[Ans: -3]

(b) $\iint_R \sqrt{4 - x^2 - y^2} \, dA$, where $R = \{(x, y) : x^2 + y^2 \leq 4, x \geq 0\}$.

[Ans: π^2]

17. Use polar coordinates to find the volume of the given solid.

(a) Under the cone $z = \sqrt{x^2 + y^2}$ and above the disk $x^2 + y^2 \leq 4$.

[Ans: 4π]

(b) Enclosed by the hyperboloid $-x^2 - y^2 + z^2 = 1$ and the plane $z = 2$.

[Ans: $14\pi/3$]

5b(iii) Triple Integrals

18. Evaluate the iterated integral.

(a) $\int_0^1 \int_x^{2x} \int_0^y 2xyz \, dz dy dx$ [Ans: 5/8]

(b) $\int_0^3 \int_0^1 \int_0^{\sqrt{1-z^2}} ze^y \, dx dz dy$ [Ans: $(e^3 - 1)/3$]

(c) $\int_0^{\sqrt{\pi}} \int_0^x \int_0^{xz} x^2 \sin y \, dy dz dx$ [Ans: $\pi^2/4 - 1$]

19. Evaluate the triple integral.

(a) $\iiint_E 2x \, dV$, where $E = \{(x, y, z) : 0 \leq y \leq 2, 0 \leq x \leq \sqrt{4 - y^2}, 0 \leq z \leq y\}$ [Ans: 4]

(b) $\iiint_E 6xy \, dV$, where E lies under the plane $z = 1 + x + y$ and above the region in the xy -plane bounded by the curves $y = \sqrt{x}$, $y = 0$ and $x = 1$. [Ans: 65/28]

(c) $\iiint_E xy \, dV$, where E is bounded by the parabolic cylinders $y = x^2$ and $x = y^2$ and the planes $z = 0$ and $z = x + y$. [Ans: 3/28]

(d) $\iiint_T x^2 \, dV$, where T is the solid tetrahedron with vertices $(0, 0, 0)$, $(1, 0, 0)$, $(0, 1, 0)$, and $(0, 0, 1)$. [Ans: 1/60]

20. Use a triple integral to find

(a) the volume of the tetrahedron enclosed by the coordinate planes and the plane $x + y + 3z = 6$. [Ans: $\frac{1}{3}(18)(2) = 12$]

(b) the volume of the tetrahedron enclosed by the planes $2x + 3y + z = 6$, $x = 3y$, $x = 0$ and $z = 0$. [Ans: $\frac{1}{3}(\frac{2 \times 2}{2})(6) = 4$]

5c(i) Change of Variables for Triple Integrals

21. A transformation is defined by the equations $x = pu$, $y = qv$, $z = rw$. Find the Jacobian of the transformation. Use the change of variable, evaluate the integral $\iiint_R dV$, where

R is the solid enclosed by the ellipsoid $\frac{x^2}{p^2} + \frac{y^2}{q^2} + \frac{z^2}{r^2} = 1$. [Ans: $4\pi pqr/3$]

22. Use the transformation $x = u^2$, $y = v^2$, $z = w^2$ to find the volume of the region bounded by the surface $\sqrt{x} + \sqrt{y} + \sqrt{z} = 1$ the coordinate planes. [Ans: 1/90]

5c(ii) Cylindrical Coordinates

23. Write the equations in cylindrical coordinates.

(a) $3x + 2y + z = 6$ [Ans: $z = 6 - r(3 \cos \theta + 2 \sin \theta)$]

(b) $-x^2 - y^2 + z^2 = 1$ [Ans: $z^2 = 1 + r^2$]

24. Use cylindrical coordinates to evaluate

(a) $\iiint_E \sqrt{x^2 + y^2} \, dV$, where E is the region that lies inside the cylinder $x^2 + y^2 = 16$ and between the planes $z = -5$ and $z = 4$. [Ans: 384π]

(b) $\iiint_E (x^2 + y^2) \, dV$ where E is the solid that lies within the cylinder $x^2 + y^2 = 1$, above the plane $z = 0$, and below the cone $z^2 = 9x^2 + 9y^2$. [Ans: $6\pi/5$]

- (c) $\iiint_E (x^2 + y^2)^{3/2} dV$, where E is the region that lies inside the cylinder $x^2 + y^2 = 4$ and between the planes $z = -1$ and $z = 3$. [Ans: $256\pi/5$]

5c(iii) Spherical Coordinates

25. Write the equation in spherical coordinates.

(a) $x^2 - 2x + y^2 + z^2 = 0$ [Ans: $\rho = 2 \cos \theta \cos \phi$]

(b) $x + 2y + 3z = 1$ [Ans: $\rho(\cos \theta \cos \phi + 2 \cos \theta \sin \phi + 3 \sin \theta) = 1$]

26. Use spherical coordinates to evaluate

(a) $\iiint_B (x^2 + y^2 + z^2)^2 dV$, where B is the ball with the origin as centre and has a radius of 5. [Ans: $312500\pi/7$]

(b) $\iiint_E x dV$, where E lies between the spheres $x^2 + y^2 + z^2 = 1$ and $x^2 + y^2 + z^2 = 9$ in the first octant. [Ans: 4]

(c) $\iiint_E x^2 dV$, where E is bounded by the xz -plane and the hemisphere $y = \sqrt{9 - x^2 - z^2}$ and $y = \sqrt{16 - x^2 - z^2}$. [Ans: $37\pi^2/12$]

(d) $\iiint_B (x^2 + y^2 + z^2)^{1/2} dV$ where $B = \{(x, y, z) : x^2 + y^2 + z^2 \leq 4\}$. [Ans: $7\pi^2$]